

Beyond Logits: Metastable Latent Dynamics for Sample-Efficient Best-of-N Selection in LLMs

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Abstract

Best-of-N selection improves reasoning in large language models (LLMs) by allocating test-time compute to sample candidate trajectories, but it relies on reliable verification. Widely used proxies have complementary failure modes: logit-confidence signals can suffer from calibration collapse, where confidence becomes misaligned with correctness, while sample agreement can be costly or brittle. Instead, we analyze the model’s latent dynamics during inference. Motivated by metastable dynamics in cognitive systems, we introduce Latent Velocity Entropy (LVE), a training-free metric that quantifies the temporal concentration of internal representation updates. Experiments on four reasoning benchmarks, AIME25, GPQA, MATH500, and BRUMO25, show that LVE-based selection remains informative when logit-confidence baselines fail, with the clearest gains on reasoning-dense mathematical tasks and more modest gains on knowledge-heavy GPQA. On MATH500, LogNorm-LVE reaches 92.2% Pass@1 at $N = 10$ and nearly matches 10-sample majority voting with only 3 samples. Cross-model, bootstrap, and length-controlled analyses indicate that LVE is not reducible to sequence length, while also clarifying the task-dependent limits of the signal. Code is available at <https://github.com/ZopenDP/LatentVelocityEntropy>.

1. Introduction

Large language models (LLMs) have achieved substantial performance on complex reasoning tasks by generating step-by-step reasoning trajectories (Wei et al., 2022; Liang et al.,

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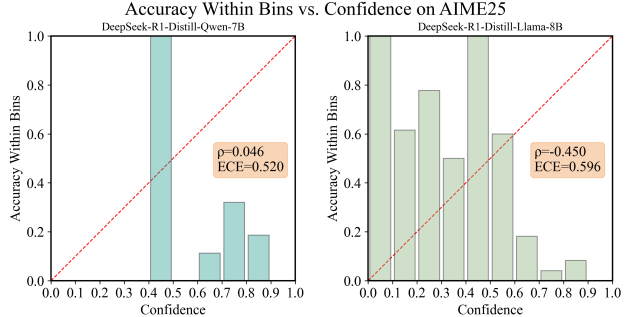


Figure 1. Reliability diagrams on AIME25 for two reasoning-capable checkpoints. Despite high output confidence (x-axis), empirical accuracy (y-axis) remains decoupled ($\rho \approx 0$) or even negatively correlated ($\rho < 0$) with self-certainty (red dashed line: ideal calibration). This illustrates that logit-derived confidence can be poorly aligned with empirical accuracy.

2022; White et al., 2024; Rein et al., 2024; Zhou et al., 2023). Building on this capability, recent research indicates that performance can be further improved through test-time inference strategies, which allocate additional computation during decoding (Wang et al., 2023; Snell et al., 2024). A cornerstone method within this paradigm is Best-of-N (BoN) selection (Cobbe et al., 2021), where the system samples multiple reasoning trajectories and selects a candidate according to a ranking or verification criterion. The efficacy of this approach is bounded by the verification challenge: to scale inference, one must reliably distinguish correct reasoning from plausible but incorrect hallucinations.

While training external Reward Models (RMs) offers a systematic verification approach (Lightman et al., 2024; Yu et al., 2025; Du et al., 2026), the substantial cost of human preference annotation and deployment latency limits their widespread applicability. Consequently, efficient inference has increasingly relied on intrinsic signals derived directly from the generating model. Prior work has shown that such signals, including logit-derived confidence proxies and cross-sample consistency, can serve as strong, training-free proxies for correctness (Kadavath et al., 2022; Wang et al., 2023; Aggarwal et al., 2023; Wang et al., 2024b; Li et al., 2024). These self-evaluation metrics have thus become a prevalent method for scalable Best-of-N selection.

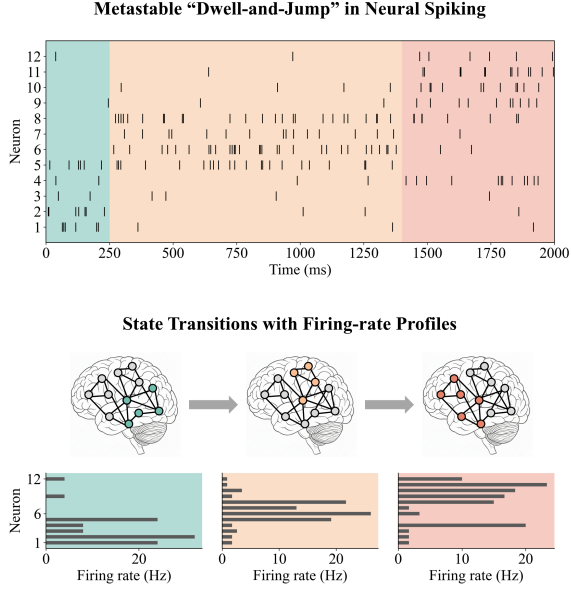


Figure 2. Metastable “dwell-and-jump” dynamics. Schematic visualization adapted from prior work on metastable population activity and state switching (Brinkman et al., 2022). *Top*: spike raster with state segmentation (colored blocks), illustrating stable dwell periods and brief transitions. *Bottom*: State transitions with firing-rate profiles over a shared population (12 neurons in total; 5/4/6 neurons predominantly active per state with overlap), highlighting selective, structured reconfiguration rather than unstructured fluctuation.

However, the efficacy of these intrinsic metrics rests on the critical assumption of calibration: that the model’s expressed confidence accurately reflects its probability of correctness. Recent studies show that this assumption can break down: LLMs may exhibit a phenomenon termed *Calibration Collapse* (Liu et al., 2025; Zhang et al., 2024; Leng et al., 2025; Li et al., 2025b; Dai et al., 2026), where confidence becomes decoupled from correctness and models maintain high log-probabilities even during hallucination. Confidence-based metrics can therefore become misleading and can even underperform random choice for Best-of-N selection, as shown in Figure 1.

To address this dilemma, we move beyond output-level signals and instead probe the model’s internal dynamics for intrinsic signatures of reasoning quality. We use cognitive metastability as a motivating prior: in biological systems, activity can alternate between relatively stable configurations (dwell periods) and intermittent transitions that incorporate new information (Rabinovich et al., 2008; Tognoli & Kelso, 2014), as illustrated in Figure 2. Rather than assuming that LLMs literally implement the same mechanism, we operationalize the measurable proxy that this prior suggests: the *temporal concentration* of latent updates during generation. Building on this, we formalize the generation process as a

latent trajectory and test whether successful reasoning trajectories concentrate latent updates into a few key junctures more strongly than flawed generations.

To capture this stage-like reconfiguration, we introduce **Latent Velocity Entropy (LVE)**, a training-free metric based on hidden-state update magnitudes. LVE quantifies the *temporal concentration* of latent updates by measuring how normalized update magnitudes deviate from uniformity: lower entropy indicates concentrated, decisive transitions, while higher entropy indicates diffuse updating.

Our contributions are as follows:

- Motivated by metastability in cognitive dynamics, we formalize update concentration as a within-trajectory signal and show that sparse latent transitions distinguish successful from unsuccessful reasoning, capturing evidence concentrated at a few key junctures rather than only in aggregate selection outcomes.
- We propose Latent Velocity Entropy (LVE) and its length-normalized variant (LogNorm-LVE) to quantify this signal without training a verifier or relying on logit-derived confidence. The normalized variant mitigates length sensitivity while preserving the same training-free scoring framework.
- We evaluate LVE on four reasoning benchmarks and several open-weight model families. The strongest gains occur on MATH500, where LogNorm-LVE reaches 92.2% Pass@1 at $N = 10$ and nearly matches 10-sample majority voting with 3 samples; bootstrap analyses characterize the reliability of these gains over datasets.

2. Method: Latent Velocity Entropy for Best-of- N Selection

We present a training-free selection method for Best-of- N that scores each candidate using the temporal organization of hidden-state updates along generation. Given N sampled candidates, we compute a per-step update-magnitude (velocity) sequence along a chosen layer, normalize it into a probability mass over steps (normalized velocity), and select the candidate with minimum Latent Velocity Entropy (LVE), optionally with length normalization and a minimal update-magnitude gate.

2.1. Problem Setup

Best-of- N Selection. We operate in the standard inference-time scaling setting using Best-of- N (BoN) selection. Given an input query x , a generator policy π_θ samples N candidate reasoning chains $\mathcal{Y} = \{y^{(1)}, \dots, y^{(N)}\}$. A selection strategy then chooses the candidate $\hat{y}$ with the

highest score under a scoring function $S(y)$:

$$\hat{y} = \operatorname{argmax}_{y^{(i)} \in \mathcal{Y}} S(y^{(i)}) \quad (1)$$

LLM Latent Trajectory. Since ground-truth labels are unavailable at test time, we seek an unsupervised proxy for $S(y)$. We fix a transformer layer ℓ and extract the residual-stream hidden state at each generation step t . For each candidate $y^{(i)}$, this yields a time-indexed sequence (trajectory) of hidden states $H^{(i)} = (h_0, h_1, \dots, h_T)$, where $h_t \in \mathbb{R}^d$ denotes the layer- ℓ representation after producing token t . Our goal is to derive a scoring function $S(H)$ based strictly on the *dynamical signature* of this trajectory, independent of logit-derived confidence proxies that are prone to calibration collapse.

2.2. Latent Dynamics as a Selection Signal

Motivation: Metastability and Dwell-and-Jump Dynamics. Cognitive neuroscience characterizes effective processing as *metastable*—a regime that balances *integration* (stability) and *segregation* (flexibility) (Rabinovich et al., 2008; Tognoli & Kelso, 2014). From this perspective, cognition does not evolve uniformly; instead, it alternates between coordinated “dwell” periods that preserve context and rapid “switching” events that integrate new information. We use this framework as a motivation for a measurable hypothesis in LLMs: useful reasoning trajectories may allocate latent update mass unevenly over time, with a small number of high-update steps embedded in longer lower-update regions.

Update concentration as a metastable signature. In the latent space, this organization corresponds to *update concentration*: a small number of tokens account for a large share of the representational change. Conversely, less selective trajectories can manifest as either *distributed* profiles (diffusive drift with nearly uniform updates) or *fixated* trajectories (stagnation). We do not treat these regimes as a complete mechanistic theory of reasoning; instead, we use them to derive an operational selection score and validate it empirically.

Definition 2.1 (Update-Time Distribution). Given a trajectory of hidden states $H = (h_0, \dots, h_T) \in (\mathbb{R}^d)^{T+1}$, define the nonnegative step velocities $v_t(H)$ by

$$v_t(H) \triangleq \|h_t - h_{t-1}\|_2, \quad t = 1, \dots, T. \quad (2)$$

Let $Z(H) \triangleq \sum_{t=1}^T v_t(H)$. Define the update-time distribution $P_H = (p_1, \dots, p_T) \in \Delta^{T-1}$ (i.e., the normalized velocity over time) by

$$p_t \triangleq \begin{cases} \frac{v_t(H)}{Z(H)} & \text{if } Z(H) > 0, \\ \frac{1}{T} & \text{if } Z(H) = 0, \end{cases} \quad t = 1, \dots, T. \quad (3)$$

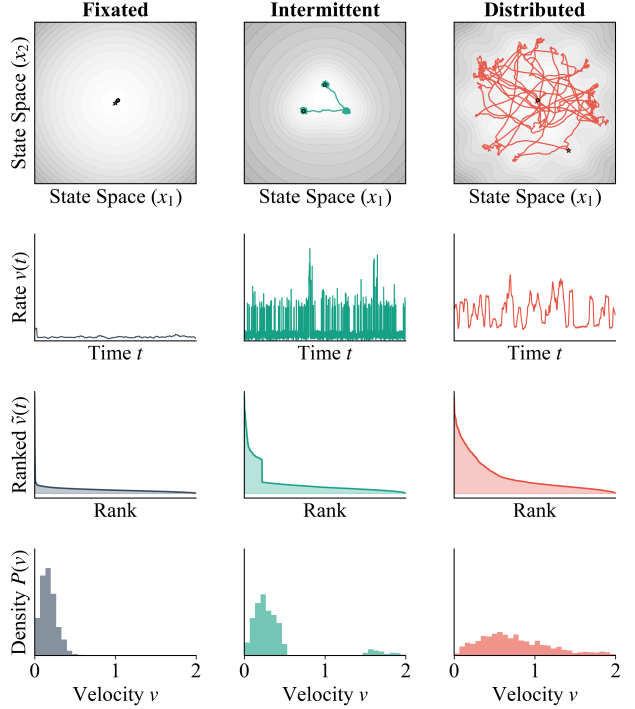


Figure 3. Conceptual schematic of latent update profiles. (Left) Fixated: near-zero movement. (Middle) Intermittent updates: a small number of decisive moments carry most update mass. (Right) Distributed updates: mass is spread broadly. Row 1: landscape potentials. Row 2: raw velocity $v(t)$ over time t . Row 3: ranked velocity profile, with normalized velocity sorted from high to low. Row 4: velocity density $P(v)$. The schematic uses synthetic dynamics only: intermittent trajectories follow a compound Poisson process with jumps $J \sim \text{LogNormal}(1, 0.5)$ and rate $\lambda = 0.05$; distributed trajectories follow a Wiener process with $\Delta h \sim \mathcal{N}(0, 0.1)$; fixated trajectories follow an attractor-return process with coefficient 0.3 and noise 0.1. LVE measures the temporal concentration of hidden-state updates while treating the biological analogy as a motivating prior.

We interpret P_H as a categorical (multinomial over indices) distribution over steps $\{1, \dots, T\}$: sampling $\tau \sim P_H$ selects a salient update step (a switching event) with probability proportional to its update magnitude (or uniformly if the trajectory is fixated).

Dynamic regimes. Figure 3 summarizes three update-magnitude regimes implied by the metastability-inspired view. We use the schematic as an operational taxonomy, not as a mechanistic account of every transformer computation. The synthetic 2D state-space potential illustrates how the same total movement can be organized differently over time:

- *Intermittent (concentrated)*: update mass is concentrated in a small number of steps. This is the intended “dwell-and-jump” signature: the model maintains a relatively stable context during dwell periods and makes

abrupt representational changes at a few transition points. In the ranked velocity profile, this appears as a sharp elbow, with a few high-magnitude updates followed by a low baseline.

- *Distributed*: update mass is spread broadly and nearly uniformly across the sequence. This corresponds to diffusion-like drift without clear reasoning milestones, producing a smooth ranked velocity decay and high entropy.
- *Fixated*: total update magnitude is negligible. Such trajectories may reflect stagnation or repetitive loops. The $Z(H) > \delta$ gate is included for this edge case, although the observed $Z(H)$ values in our experiments are far from zero.

LVE favors the intermittent regime because it assigns low entropy to trajectories whose latent updates are temporally selective while penalizing diffuse update allocation.

Distinction from Output-Level Metrics. This perspective differs from prior work on Uniform Information Density (UID) (Gwak et al., 2025). While UID also links quality to non-uniformity, it operates on *output token surprisal*, which is vulnerable to calibration collapse: a model can be confidently wrong, producing low, uniform surprisal even during hallucination. In contrast, we target the *latent dynamics* of the generation process. By measuring the concentration of internal updates rather than surface-level prediction probabilities, we obtain a complementary trajectory-derived signal that can remain informative when output confidence is miscalibrated.

2.3. Latent Velocity Entropy (LVE)

We propose LVE as a training-free selection criterion that operationalizes this metastability-inspired *update concentration* signature using internal dynamics rather than logit-derived confidence proxies. Concretely, we measure the entropy of the update-time distribution P_H defined in Section 2.2, which quantifies the *temporal concentration* (deviation from uniform allocation) of latent updates.

Definition 2.2 (Latent Velocity Entropy (LVE)). The Latent Velocity Entropy is the Shannon entropy of the update-time distribution P_H :

$$\mathcal{H}_{vel}(H) \triangleq - \sum_{t=1}^T p_t \log p_t, \quad \text{with } 0 \log 0 \triangleq 0. \quad (4)$$

Interpretation. Low $\mathcal{H}_{vel}(H)$ indicates that the update-time distribution P_H deviates from the uniform distribution U_T . In particular,

$$D_{KL}(P_H \| U_T) = -\mathcal{H}_{vel}(H) + \log T. \quad (5)$$

We therefore use LVE as a proxy for rejecting trajectories with near-uniform, non-selective update allocation over time. This is a test-time heuristic, and we empirically validate its separability and selection performance in Section 3.

Selection Criterion. Given N candidates, we score each by $\mathcal{H}_{vel}(H_y)$ and select the minimum-entropy trajectory among those with sufficient total update magnitude:

$$\hat{y} = \operatorname{argmin}_{y \in \mathcal{Y}} \mathcal{H}_{vel}(H_y) \quad \text{s.t.} \quad Z(H_y) > \delta \quad (6)$$

where δ is a small threshold to filter near-fixated trajectories. Notably, since total update magnitudes in LLM reasoning are typically well above zero, this constraint is naturally satisfied by any substantive generation. In our experiments, observed $Z(H)$ values are far from zero (about 2.38×10^4 for Qwen and 3.4×10^3 for Llama in the profiled settings), so the gate is effectively inactive and LVE behaves as a parameter-free score in practice.

Length Normalization. The theoretical upper bound of LVE is $\log T$, meaning that longer trajectories inherently possess a larger capacity for entropy. In the context of Best-of- N selection where candidate lengths vary, comparing raw entropy values biases selection against longer, potentially valid chains. To correct for this, we adopt *LogNorm-LVE*:

$$\bar{\mathcal{H}}_{vel}(H) \triangleq \frac{\mathcal{H}_{vel}(H)}{\log T}, \quad T > 1 \quad (7)$$

This normalized entropy is bounded in $[0, 1]$ and measures relative diffusivity, or closeness to a uniform allocation of update mass. Lower values indicate stronger deviation from uniformity; equivalently, $1 - \bar{\mathcal{H}}_{vel}(H)$ is the normalized KL divergence from the uniform allocation. Degenerate trajectories with at most one velocity step are excluded or assigned a fallback score in implementation. Raw LVE is not length-invariant because its upper bound grows with $\log T$; LogNorm-LVE is therefore our preferred variant when candidates have substantially different lengths. Section 3 and Appendix D.5 further quantify the remaining relationship between LVE and sequence length.

3. Experiments

3.1. Experimental Setup

We evaluate unsupervised Best-of- N selection across multiple reasoning benchmarks (AIME25, BRUMO25, MATH500, GPQA) and perform candidate-level ranking analysis on MATH500. We use the vLLM framework for high-throughput generation and HuggingFace Transformers for hidden state extraction.

Models. The main experiments use DeepSeek-R1-Distill-Qwen-7B (28 layers) and DeepSeek-R1-Distill-Llama-8B

Table 1. Comparison of selection methods across datasets at $N = 10$. Each dataset cell reports Pass@1 accuracy (%) with a 95% bootstrap confidence interval; Avg. is the arithmetic mean across datasets. Wide intervals on AIME25/BRUMO25 reflect their small evaluation size, so several differences fall within statistical uncertainty; MATH500 provides tighter separation. Best point estimates are in **bold**.

Method	DeepSeek-R1-Distill-Qwen-7B					DeepSeek-R1-Distill-Llama-8B				
	AIME25	BRUMO25	MATH500	GPQA	Avg.	AIME25	BRUMO25	MATH500	GPQA	Avg.
Random	23.3 [10.0, 40.0]	23.3 [10.0, 40.0]	81.4 [78.0, 84.8]	49.5 [42.4, 56.6]	44.4	23.3 [10.0, 40.0]	16.7 [3.3, 30.0]	76.4 [72.6, 80.0]	47.5 [40.4, 54.5]	41.0
Self-Certainty	20.0 [6.7, 36.7]	10.0 [0.0, 20.0]	70.2 [66.2, 74.2]	52.0 [44.9, 58.6]	38.1	6.7 [0.0, 16.7]	10.0 [0.0, 23.3]	60.6 [56.4, 64.8]	46.5 [39.4, 53.5]	30.9
DeepConf	16.7 [3.3, 30.0]	13.3 [3.3, 26.7]	72.6 [68.6, 76.6]	54.0 [47.0, 60.6]	39.2	10.0 [0.0, 23.3]	16.7 [3.3, 30.0]	73.6 [69.6, 77.4]	53.0 [46.0, 60.1]	38.3
Self-Consistency	30.0 [13.3, 46.7]	46.7 [30.0, 63.3]	87.6 [84.6, 90.4]	52.5 [45.5, 59.6]	54.2	23.3 [10.0, 40.0]	26.7 [13.3, 43.3]	83.8 [80.6, 87.0]	54.0 [47.0, 61.1]	47.0
Low-Token-Entropy	26.7 [10.0, 43.3]	43.3 [26.7, 60.0]	86.4 [83.2, 89.2]	53.0 [46.5, 59.6]	52.4	16.7 [3.3, 30.0]	20.0 [6.7, 36.7]	83.0 [79.6, 86.2]	56.6 [49.5, 63.1]	44.1
UID-Global	30.0 [13.3, 46.7]	43.3 [26.7, 60.0]	90.6 [87.8, 93.0]	52.0 [44.9, 59.1]	54.0	30.0 [13.3, 46.7]	16.7 [3.3, 30.0]	83.6 [80.2, 86.8]	52.0 [44.9, 59.1]	45.6
UID-Local	30.0 [13.3, 46.7]	50.0 [33.3, 66.7]	90.0 [87.4, 92.6]	50.5 [43.4, 57.6]	55.1	33.3 [16.7, 50.0]	33.3 [16.7, 50.0]	84.0 [80.8, 87.2]	51.0 [43.9, 58.1]	50.4
LVE (Ours)	30.0 [13.3, 46.7]	50.0 [33.3, 66.7]	90.4 [87.8, 92.8]	55.6 [48.5, 62.6]	56.5	33.3 [16.7, 50.0]	36.7 [20.0, 53.3]	85.4 [82.2, 88.4]	54.5 [47.5, 61.1]	52.5
LogNorm-LVE (Ours)	30.0 [13.3, 46.7]	50.0 [33.3, 66.7]	92.2 [89.8, 94.4]	53.5 [46.5, 60.1]	56.4	33.3 [16.7, 50.0]	40.0 [23.3, 56.7]	85.0 [81.8, 88.0]	53.0 [46.0, 60.1]	52.8

(32 layers). We abbreviate these models as DeepSeek-R1-Qwen and DeepSeek-R1-Llama in figures and extended tables. To assess whether the signal is specific to these distilled reasoning checkpoints, Appendix D.4 extends the evaluation to Gemma-3-4B-It and Qwen3-8B.

Sampling. For each problem, we sample $N \in \{3, 5, 10, 15\}$ distinct reasoning chains with temperature 0.7 and $\text{top}_p = 0.9$.

Baselines. *Random* uniformly samples one candidate from the N generated rationales. *Self-Consistency* (Majority Voting) (Wang et al., 2023) selects the answer by aggregating over multiple sampled candidates. *Self-Certainty* (Kang et al., 2025) estimates distributional certainty from next-token predictive distributions. *DeepConf* (Fu et al., 2025) combines token-level confidence proxies with a windowed operator to capture intermediate breakdowns. *Low-Token-Entropy* (Kadavath et al., 2022) ranks candidates by the mean entropy of next-token distributions. *UID-Global / UID-Local* (Gwak et al., 2025) compute information density over response steps separated by $\backslash n \backslash n$, measuring global variance and local irregularity across steps.

Our Method. Latent Velocity Entropy (LVE) and LogNorm-LVE score each candidate by the temporal concentration of hidden-state updates; Section 2 provides the formal definitions and theoretical motivation.

Metrics. The primary metric is the *Pass@1 Accuracy* of the selected response. Table 1 reports accuracy at $N = 10$, and Figure 4 reports candidate-level ROC/AUC for score separability.

For each candidate rationale y , a method computes an unsupervised score $S(y)$ and selects $\hat{y} = \arg\max_y S(y)$ (or $\arg\min$ for entropy-based metrics).

3.2. Main Results

As shown in Table 1 and Figure 4, our experimental evaluation yields several primary findings regarding calibration failure, task-dependent selection performance, model diversity, and the empirical update-concentration signal.

Calibration Collapse in Distilled Models. We first observe that confidence-based methods underperform on the distilled DeepSeek-R1 models. The *Self-Certainty* baseline, which relies on output token probabilities, falls below random selection in the Qwen aggregate (38.1% vs. 44.4% avg.) and degrades sharply on MATH500. This supports the calibration-collapse hypothesis: output probability can become weakly or negatively aligned with factual correctness.

Task-Dependent Gains from Latent Dynamics. LVE-based selection performs best on reasoning-dense mathematical tasks. On MATH500, LogNorm-LVE achieves 92.2% Pass@1 accuracy on Qwen, above Self-Consistency (87.6%) and UID-Local (90.0%) with a tighter bootstrap interval. On AIME25 and BRUMO25, wide confidence intervals make many apparent margins uncertain. On GPQA, gains are more modest and sometimes baseline-dependent, suggesting that update concentration is most useful when correctness depends on discovering a structured reasoning trajectory rather than retrieving factual knowledge.

Separability Analysis (ROC). Figure 4 helps explain the MATH500 behavior. With an AUC of 0.89 on Qwen and

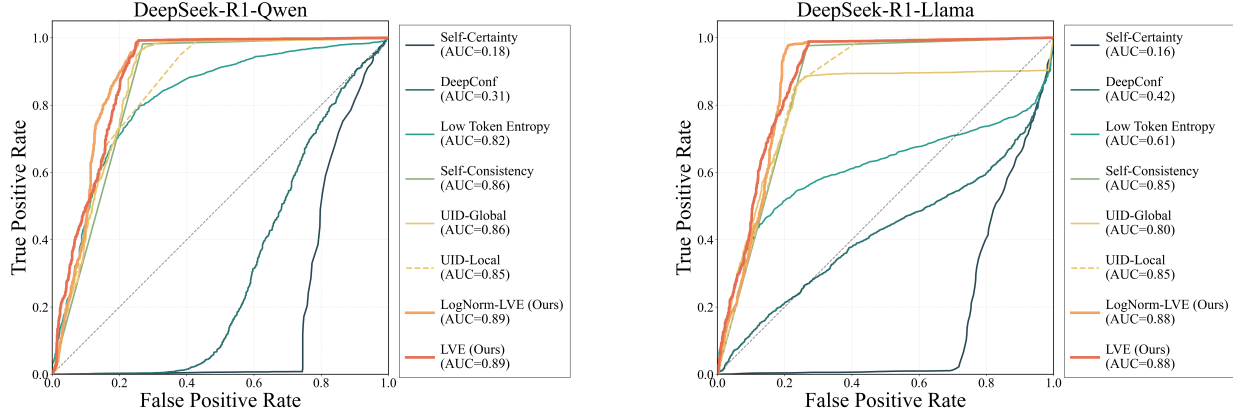


Figure 4. Candidate-level ROC on MATH500 (500 problems). We evaluate how well each unsupervised score *rank*s correct vs. incorrect candidates (positive class: correct). Curves closer to the top-left indicate better separability. For scores where *lower is better* (e.g., LVE / LogNorm-LVE, entropy-based metrics), we negate the score so that larger values indicate higher correctness. LVE and LogNorm-LVE achieve the highest AUC among the tested methods on both models (Qwen: AUC=0.89; Llama: AUC=0.88), while log-probability-based Self-Certainty performs poorly under calibration collapse (AUC=0.18/0.16).

0.88 on Llama, LVE separates correct from incorrect candidates better than the tested output-entropy baselines. In contrast, Self-Certainty has a low AUC of 0.18/0.16, meaning the model is often more confident when it is wrong. Thus, internal update concentration provides a useful complementary signal when surface confidence is unreliable.

Model Diversity. Appendix D.4 extends the evaluation to Gemma-3-4B-It and Qwen3-8B across 32 settings. Either LVE or LogNorm-LVE appears among the methods attaining one of the top three distinct point-estimate levels in 27/32 settings. These results indicate that the signal is not restricted to the original DeepSeek-R1 distilled checkpoints, while also reinforcing the task-dependent pattern: performance is strongest on mathematical reasoning and less decisive on GPQA.

Empirical Dynamics. We interpret the performance pattern as empirical support for the update-concentration view. Figure 4, the visualization in Figure 6, and the Top- k velocity-mass analysis in Appendix D.6 all suggest that correct reasoning trajectories more often concentrate latent update mass into sparse transitions than incorrect trajectories.

3.3. Sample Efficiency Analysis

We evaluated selection accuracy as a function of sample count $N \in \{3, 5, 10, 15\}$ in Figure 5. Key findings regarding DeepSeek-R1-Qwen on MATH500:

- At $N = 3$, LVE achieves 87.4% accuracy, outperforming Self-Consistency (84.2%) by 3.2 percentage

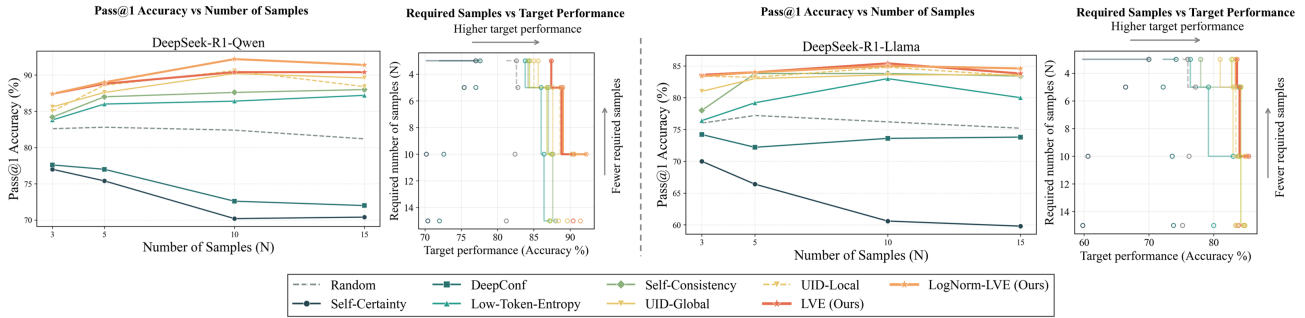


Figure 5. Sample Efficiency on MATH500. Selection accuracy as a function of sample count N on MATH500 for DeepSeek-R1-Qwen and DeepSeek-R1-Llama. LVE/LogNorm-LVE (red/orange) are competitive across both backbones and reach a peak of 92.2% at $N = 10$ on Qwen. Self-Certainty (dark blue) performs poorly under calibration collapse. On Qwen, LVE at $N = 3$ nearly matches Self-Consistency at $N = 10$. The right panel plots required samples versus target performance, where each curve reports the minimum number of samples needed to reach a given accuracy threshold; lower curves at a given target indicate better sample efficiency.

points—a nontrivial margin when computational budgets are constrained.

- LogNorm-LVE maintains a consistent point-estimate lead across all sample sizes: +2.0% at $N = 5$ (89.0% vs. 87.0%), +4.6% at $N = 10$ (92.2% vs. 87.6%), and +3.4% at $N = 15$ (91.4% vs. 88.0%).
- Self-Certainty performs *worse* than random selection across all N values (e.g., 70.2% vs. 81.4% at $N = 10$), showing that confidence-based metrics can be unreliable in this calibration-collapse setting.

These results have practical implications for MATH500-style reasoning workloads: at $N = 3$, LVE already achieves 87.4% accuracy—nearly matching Self-Consistency at $N = 10$ (87.6%)—using about 70% fewer generated candidates for this benchmark.

From the update-concentration viewpoint, this efficiency is plausible because the signal is computed within each candidate trajectory. A single candidate can exhibit low-entropy dynamics, whereas majority voting needs multiple samples to compensate for unreliable output-confidence proxies under calibration failure.

3.4. Length Control and System Overhead

Length Dependence. LVE is correlated with sequence length, so we quantify that dependence explicitly rather than treating the raw score as length-invariant. On MATH500 with DeepSeek-R1-Distill-Qwen-7B, raw LVE has Pearson correlation 0.384 with token length and 0.389 with log-token-length, but it remains discriminative after length control. In a 5-fold GroupKFold setting, length alone yields AUROC 0.517, while adding raw LVE raises AUROC to 0.626. The detailed comparison against shortest-length, including the per-backbone gains and disagreement rate, is reported in Appendix D.5; overall, these controls support LogNorm-LVE as the preferred variant when candidate lengths vary.

Scoring Overhead. Our implementation decouples optimized vLLM generation from LVE scoring: candidates are generated with vLLM, then replayed in a separate batched GPU forward pass to extract hidden-state trajectories. On MATH500 with DeepSeek-R1-Distill-Llama-8B, $N = 10$, bfloat16, and 4 RTX 3090 GPUs, this scoring stage takes 870 ms versus 1257 ms for the profiled logits-based scoring pipeline, and transfers 7.1 KB/token rather than 608 KB/token. This corresponds to 30.85% lower scoring latency and 98.8% lower transfer payload in the measured setup.

3.5. Visual Analysis: The Velocity Landscape

To inspect the update-concentration signal behind LVE, we visualize the latent trajectories of generated responses. Figure 6 offers two complementary views: a 2D projection of the latent state trajectories (top) and temporal velocity profiles (bottom).

Latent State Trajectories (Top Row). The background heatmap provides a qualitative 2D view of trajectory organization, with darker regions indicating lower projected movement density under the visualization procedure. The correct generation (Left) shows a metastability-like dwell-and-jump pattern: the trajectory remains in compact projected regions for extended periods (purple segments) before executing rapid, high-velocity transitions (yellow spikes). This pattern is consistent with step-like reasoning transitions. In contrast, the incorrect generation (Right) displays a more *diffusive* pattern: the trajectory wanders continuously with persistent, medium-amplitude fluctuations (teal/green segments), without forming similarly compact projected segments.

Temporal Velocity Profiles (Bottom Row). The velocity mass plots illustrate the same temporal structure: the correct response concentrates more of its “velocity budget” into a few sparse peaks, resulting in lower entropy, while the incorrect response spreads velocity mass more uniformly across time. This qualitative case study is complemented by the dataset-level analyses in Figure 4 and Appendix D.6.

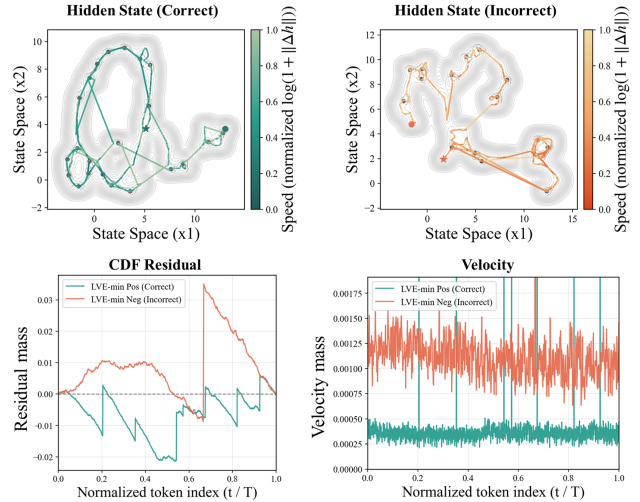


Figure 6. Latent-state dynamics and velocity-based diagnostics. Upper row: 2D projected latent-state trajectories with a qualitative background visualization of trajectory organization; trajectories are colored by normalized $\log(1 + \|\Delta h\|)$. (Left: correct prediction. Right: incorrect prediction.) Lower row: velocity-based diagnostics over the length-normalized token index (t/T). (Left: CDF residual, deviations from a uniform accumulation. Right: non-cumulative velocity mass.)

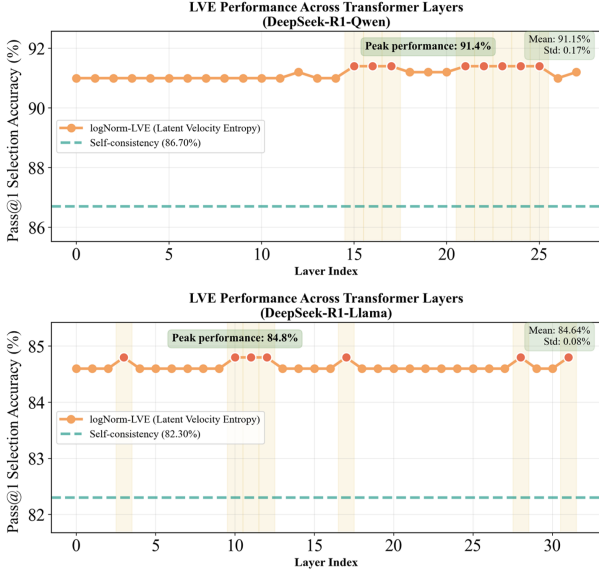


Figure 7. Layer-wise LogNorm-LVE performance across transformer depths. Pass@1 accuracy stays competitive across broad layer ranges on both DeepSeek-R1-Qwen and DeepSeek-R1-Llama, indicating that selection does not appear to rely on a single chosen representation depth.

We provide a detailed analysis of representative failure modes and their internal dynamics in Appendix D.8.

3.6. Layer Sensitivity

To study the sensitivity of LogNorm-LVE to representation depth, we evaluate its performance using latent features extracted from each transformer layer. Figure 7 reports the layer-wise top-1 selection accuracy for two representative backbones, DeepSeek-R1-Qwen and DeepSeek-R1-Llama.

Across both models, LogNorm-LVE remains above the corresponding Self-Consistency baseline across many evaluated layers. Its layer-wise mean is substantially higher than the baseline and its standard deviation remains small, suggesting robustness to representation depth.

Although peak performance occurs in certain layers (highlighted in the shaded regions), the overall curve is relatively flat. This indicates limited sensitivity to layer tuning and suggests that the update-concentration signal used by LogNorm-LVE is preserved across transformer layers.

We present the layer-wise robustness analysis on one representative dataset in the main text, while results on additional datasets are provided in Appendix D.7.

4. Related Work

Best-of- N Selection

Best-of- N (BoN) allocates inference compute across multiple stochastic reasoning trajectories and selects a final output via aggregation or ranking, trading test-time budget for improved robustness and accuracy (Cobbe et al., 2021; Wang et al., 2023). Early Best-of- N frameworks often relied on external evaluators, such as reward or verifier models, to select among sampled candidates (Lightman et al., 2024), incurring substantial computational and deployment overhead. More recent work has therefore focused on improving the efficiency and practicality of the selection stage by exploiting signals available during or after generation, without introducing separate external evaluators. We organize these selection strategies according to the type of signal they leverage, broadly categorizing them into output-level signals and internal-level signals.

Output-Level Signals

Majority voting. Self-consistency (Wang et al., 2023) selects the most frequent answer among sampled outputs and is effective for discrete, comparable answers. This approach has been extended in several directions: (1) handling open-ended text (e.g., Choi & Li (2026) via graph clustering over pairwise text similarity); (2) softening exact-match voting with likelihood-based continuous criteria (Wang et al., 2024b); and (3) reducing computational cost through adaptive or early-stopping schemes. These cost-reduction methods dynamically allocate sampling budgets or halt sampling once sufficient agreement is observed (Aggarwal et al., 2023; Li et al., 2024). In contrast, MoB (Rakhsha et al., 2025) improves robustness under imperfect reward models by bootstrapping the distribution of BoN outputs.

Logit-based signals. Another line ranks or prunes candidates using token-distribution statistics such as log-likelihood, entropy, or other confidence-based measures (Kang et al., 2025; Fu et al., 2025), enabling filtering and early stopping without hidden-state access, but implicitly assumes output confidence correlates with reasoning validity, an assumption shown to be brittle in open-ended reasoning settings, where probabilistic metrics can remain high despite logical inconsistencies and primarily reflect surface-level fluency or in-distribution priors (Kim & Kim, 2026).

Internal-Level Signals

Causal analyses of reasoning indicate that logical inference operates over structured, dependent latent variables, tying reasoning quality to the evolution and interaction of latent representations rather than surface outputs alone (Deng et al., 2026). This matters when similar final answers arise from different internal trajectories.

Prior internal-signal methods can be broadly grouped into three categories: (i) state-centric statistics (Chen et al., 2025; Wang et al., 2025a), which leverage activation sparsity or cross-sample agreement to select trajectories; (ii)

information-flow metrics (Gwak et al., 2025; Vilas et al., 2025), which characterize reasoning quality via step-wise or trajectory-level distributional signals, such as entropy or latent-trajectory change patterns; and (iii) intrinsic evaluators or latent supervision mechanisms, including lightweight verifiers trained on hidden states for trajectory ranking (Yu et al., 2025), latent-space reward modeling approaches that optimize internal reasoning processes (Du et al., 2026), and output-free self-evaluation via progressive hidden-state chains (Wang et al., 2025b). These approaches differ in supervision and access requirements, but share the premise that internal computation can reveal selection cues not captured by the final answer alone.

Recent mechanistic studies also suggest that reasoning is associated with structured internal transitions rather than uniform token-by-token progress. Attention-based analyses identify preplan-and-anchor rhythms in which long-range contextual reference is followed by semantically influential anchor tokens (Li et al., 2025a). Complementary work on implicit reasoning and grokking links generalization to the emergence of structured circuits or pathways during training (Wang et al., 2024a; Yao et al., 2026; Li et al., 2026). These studies motivate our focus on temporal structure; we use LVE as a lightweight test-time signal for reranking sampled trajectories rather than as a circuit-level explanation.

Despite their effectiveness, many existing approaches ultimately reduce reasoning trajectories to a limited set of per-step statistics or aggregated scores. While recent work has begun to leverage step-wise or trajectory-level temporal signals, explicitly modeling fine-grained latent state transition dynamics as the primary object of selection remains comparatively under-explored.

5. Discussion and Conclusion

Our empirical results suggest that latent trajectory structure is a useful test-time signal for Best-of- N selection. LVE improves selection most clearly when output confidence is poorly calibrated and candidate quality depends on a structured reasoning path rather than surface fluency alone. In this setting, temporally concentrated hidden-state updates provide information that is not directly visible from the final answer or next-token probabilities.

The task-level pattern is important. On MATH500, LVE and LogNorm-LVE provide the clearest gains, consistent with the hypothesis that mathematical reasoning often requires sparse internal transitions between intermediate states. On knowledge-heavy GPQA, the evidence is weaker and more baseline-dependent, suggesting that update concentration is most informative when correctness depends on discovering and maintaining a reasoning trajectory. Small benchmarks such as AIME25 and BRUMO25 further produce wide con-

fidence intervals, so close point-estimate differences should be interpreted cautiously. This suggests that the signal is structured rather than universally transferable.

Against that backdrop, the comparison with baselines clarifies why the signal is complementary. Self-Consistency remains useful when many samples converge to a normalized answer, but it relies on cross-sample agreement and can miss a correct minority trajectory. UID-style statistics capture non-uniformity in output surprisal, yet their signal still depends on token-probability behavior. LVE instead scores the temporal organization of each candidate’s hidden-state path, making it useful when vote counts or surface confidence provide an incomplete view of candidate quality.

The robustness analyses refine the intended scope of this conclusion. The model-diversity study shows that LVE variants are not restricted to the original DeepSeek-R1 distilled checkpoints, but the exact ranking of methods varies across model families, sample budgets, and datasets. The length-control analysis gives a second boundary condition: raw LVE carries measurable length dependence, while LogNorm-LVE provides the preferred comparison when candidate lengths vary. Thus, confidence intervals, model diversity, and length controls are central to identifying where the trajectory cue is reliable, and where it should be treated as only one cue among several.

From a deployment perspective, LVE is best viewed as an unsupervised reranking step for open-weight models where hidden states can be replayed after generation. Conceptually, it sits between output-only voting and trained verifier approaches, using no labels or additional supervision but exploiting trajectory information unavailable from final answers alone. It is not a standalone correctness certificate: a low-entropy trajectory can indicate structured internal progress, but it does not prove that the final answer is correct. The appropriate use is comparative, as an additional trajectory-level cue that can be combined with voting, symbolic checks, or verifier scores when such signals are available. This also makes the method most natural for settings where several plausible candidates are already generated and the remaining problem is to choose among them without introducing a task-specific trained judge.

Limitations and Future Directions. LVE requires white-box access to hidden states, limiting use with closed-source models. Its gains are task-dependent and strongest on deductive reasoning benchmarks, while raw LVE remains length-correlated and therefore benefits from length normalization and explicit controls when candidate lengths vary. Future work should test whether update concentration remains informative in larger open-weight models and non-Transformer architectures such as Mamba (Gu & Dao, 2024), where the latent update geometry may differ.

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Impact Statement

This paper presents work to advance machine learning by improving the reliability and sample efficiency of large language models (LLMs) through latent dynamics analysis. Our proposed method, Latent Velocity Entropy (LVE), is a training-free selection mechanism that can reduce incorrect outputs in the evaluated reasoning-selection settings.

Potential positive impacts include improved reliability where accurate reasoning is valuable. However, like other methods that improve LLM capabilities, it could aid malicious workflows. We identify no specific negative ethical impacts beyond the general dual-use nature of AI.

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A. Theory Details

A.1. Relationship to KL Divergence from Uniform

For any trajectory with $Z > 0$, the Kullback–Leibler divergence to the diffusive (uniform) prior U_T is

$$D_{\text{KL}}(P_H \| U_T) = \sum_{t=1}^T p_t \log \frac{p_t}{1/T} \quad (8)$$

$$= \sum_{t=1}^T p_t \log p_t + \log T \sum_{t=1}^T p_t \quad (9)$$

$$= \sum_{t=1}^T p_t \log p_t + \log T \quad (10)$$

$$= -\mathcal{H}_{\text{vel}}(H) + \log T. \quad (11)$$

For fixed response length T , maximizing $D_{\text{KL}}(P_H \| U_T)$ is equivalent to minimizing $\mathcal{H}_{\text{vel}}(H)$.

A.2. Length Normalization Rationale and Ablation

Raw LVE has an intrinsic upper bound $\log T$, so longer candidates have a larger capacity for entropy even when their latent updates are comparably sparse. Normalizing by $\log T$ yields a bounded quantity that compares *relative* non-diffusivity across different lengths, aligning with the argument in Section 2.3.

Across the length-normalization ablation, LogNorm-LVE is often the strongest or tied-best variant and gives the best $N = 10$ result on Qwen, consistent with the length-capacity argument above: dividing by $\log T$ corrects the structural bias against longer candidates without over-penalizing length as linear normalization does.

A.3. Schematic Simulation Details

Figure 3 in the main text is a conceptual simulation rather than a measured LLM trajectory. The intermittent regime follows a compound Poisson process with jumps $J \sim \text{LogNormal}(1, 0.5)$ and rate $\lambda = 0.05$; the distributed regime follows a Wiener process with $\Delta h \sim \mathcal{N}(0, 0.1)$; and the fixated regime follows an attractor-return process with coefficient 0.3 and noise 0.1.

B. Experimental Protocol & Reproducibility

B.1. Datasets and Evaluation Sizes

We report Best-of- N selection results on four reasoning benchmarks, using the following evaluation sizes: AIME25 (30 problems), BRUMO25 (30 problems), GPQA (198 problems), and MATH500 (500 problems). These sizes correspond to the evaluation subsets used in our released result summaries.

Table 2. Length normalization ablation on MATH500 (500 problems). We compare raw LVE, linear normalization (Norm-LVE; dividing by T), and logarithmic normalization (LogNorm-LVE; dividing by $\log T$). Results are selection accuracy (%).

Method	$N = 3$	$N = 5$	$N = 10$	$N = 15$
DeepSeek-R1-Qwen				
LVE	87.4	88.8	90.4	90.4
Norm-LVE	85.2	86.6	89.6	89.2
LogNorm-LVE	87.4	89.0	92.2	91.4
DeepSeek-R1-Llama				
LVE	83.6	84.0	85.4	83.8
Norm-LVE	79.6	80.4	82.2	81.6
LogNorm-LVE	83.4	84.0	85.0	84.6

B.2. Prompting and Decoding

Generation follows the standard Best-of- N protocol. For each problem, we sample $N \in \{3, 5, 10, 15\}$ candidates independently using temperature 0.7 and top- $p = 0.9$. All experiments were run on a single node with four NVIDIA RTX 3090 GPUs. We use a fixed chat-style system instruction with the exact system prompt:

```
You are a helpful assistant. Please think step by step.
At the end of your solution, you MUST put the final answer inside \boxed{{}}.
```

B.3. Scoring Pipeline and Overhead

Generation and LVE scoring are decoupled. We first generate candidates with vLLM’s optimized decoding engine, then replay the prompt-response pairs in a separate batched GPU forward pass to extract hidden-state trajectories. This avoids per-token CPU-GPU synchronization during generation and does not require transferring full vocabulary-dimensional logits for LVE scoring. In the profiled setting (DeepSeek-R1-Distill-Llama-8B, MATH500, $N = 10$, bfloat16, four RTX 3090 GPUs), LVE scoring takes 870 ms versus 1257 ms for the profiled logits-based scoring pipeline. The transferred payload is 7.1 KB/token for hidden-state velocity scoring versus 608 KB/token for logits, corresponding to 30.85% lower scoring latency and 98.8% lower transfer payload in this setup.

B.4. Correctness Judging and Answer Extraction

Correctness is evaluated using the official `Qwen_Math` answer parsing and grading utilities, following Yang et al. (2024), which robustly handle LaTeX-formatted mathematical expressions and normalize equivalent answers. In brief, final answers are extracted by prioritizing the boxed expression (when present) and then graded via equivalence checking against the reference solution. For non-mathematical multiple-choice formats, the grader falls back to matching the choice marker and/or its normalized content. This protocol mirrors the common evaluation practice for math reasoning benchmarks and is designed to minimize formatting-induced false negatives.

B.5. Bootstrap Confidence Intervals

For each dataset, model, sampling budget N , and selection method, we treat each problem-level outcome as a Bernoulli variable indicating whether the selected answer is correct. We run 10,000 bootstrap trials. In each trial, we resample the evaluation problems with replacement using the original dataset size M , and recompute accuracy as the mean correctness over the resampled set. The reported 95% confidence interval is the percentile interval of these bootstrap accuracies, given by the 2.5th and 97.5th percentiles. These intervals quantify uncertainty over evaluation problems conditional on the generated candidate sets, and do not include variance from resampling generation seeds.

B.6. Candidate-Level ROC/AUC Protocol

For candidate-level separability analysis, each candidate is treated as an individual example with a binary label (correct/incorrect). For each unsupervised score, we compute an ROC curve and AUC. For scores where *lower is better* (e.g., LVE, LogNorm-LVE, entropy-based metrics), we negate the score so that larger values consistently indicate higher correctness. For Self-Consistency, a candidate receives a score of 1 if its extracted final answer matches the majority answer among candidates for the same problem, and 0 otherwise.

B.7. Reliability Diagram and ECE

We construct reliability diagrams using fixed-width bins over confidence in $[0, 1]$ with 10 bins. For each bin, we report *accuracy within bin* and the *mean confidence within bin*. Expected Calibration Error (ECE) is computed as the weighted average of the absolute bin-wise gap:

$$\text{ECE} \triangleq \sum_{b=1}^B \frac{|S_b|}{|S|} |\text{acc}(S_b) - \text{conf}(S_b)|, \quad (12)$$

where S is the full set of evaluated candidate responses, $S_b \subseteq S$ is the subset whose confidence falls into bin b , and $B = 10$.

C. Baseline Definitions and Hyperparameters

We compare against the following unsupervised BoN selection baselines: Random, Self-Consistency (Wang et al., 2023), Self-Certainty (Kang et al., 2025), DeepConf (Fu et al., 2025), Low-Token-Entropy, and UID-Global/UID-Local (Gwak et al., 2025). Unless stated otherwise, each metric is computed on the *response tokens* of a candidate using a teacher-forced forward pass (prompt + response) to obtain logits (and, for LVE, hidden states).

Notation. Let $p_t(\cdot)$ denote the next-token distribution at generation step t over vocabulary $\mathcal{V}$ of size V . Let x_t be the generated token at step t and let the response length be L tokens.

Random. Uniformly sample one candidate from the N generated candidates.

Self-Consistency (Majority Voting) (Wang et al., 2023). Extract a final answer $a^{(i)}$ from each candidate (prioritizing boxed answers when present), then select the majority answer

$$\hat{a} = \operatorname{argmax}_a \sum_{i=1}^N \mathbf{1}[a^{(i)} = a], \quad (13)$$

and return the first candidate whose extracted answer equals $\hat{a}$ (ties broken by first occurrence).

Self-Certainty (Kang et al., 2025). Compute distributional certainty from next-token predictive distributions as the (negative) cross-entropy to a uniform prior (up to a constant), averaged over response tokens:

$$\text{SC}(y) = \frac{1}{L} \sum_{t=1}^L \left(-\frac{1}{V} \sum_{x \in \mathcal{V}} \log p_t(x) - \log V \right). \quad (14)$$

Select $\hat{y} = \operatorname{argmax}_y \text{SC}(y)$.

DeepConf (Fu et al., 2025). Define a token-level uncertainty proxy using top- k log-probabilities (we use $k = 20$):

$$u_t^{(k)} = -\frac{1}{k} \sum_{j \in \text{TopK}(p_t)} \log p_t(j). \quad (15)$$

Aggregate locally with a sliding window of size 2048 over response tokens, using the minimum window-average:

$$\text{DeepConf}_{\text{raw}}(y) = \min_{w \in \mathcal{W}} \frac{1}{|w|} \sum_{t \in w} u_t^{(k)}. \quad (16)$$

We use the score $S(y) = -\text{DeepConf}_{\text{raw}}(y)$ so that larger values indicate higher confidence, and select with argmax . For diagnostic analysis, we also compute a tail statistic on the last 2048 tokens.

Low-Token-Entropy. Compute the mean next-token entropy over response tokens:

$$H(p_t) = -\sum_{x \in \mathcal{V}} p_t(x) \log p_t(x), \quad \text{Ent}(y) = \frac{1}{L} \sum_{t=1}^L H(p_t), \quad (17)$$

and select $\hat{y} = \operatorname{argmin}_y \text{Ent}(y)$.

UID-Global / UID-Local (Gwak et al., 2025). Segment the decoded response into steps separated by `\n\n`, yielding $s_1, \dots, s_M$. Define step-wise information density from token surprisal:

$$\text{ID}(s_m) = \frac{1}{|s_m|} \sum_{t \in s_m} (-\log p_t(x_t)), \quad (18)$$

Table 3. Full $N = 10$ comparison with 95% bootstrap confidence intervals. Each dataset cell reports Pass@1 accuracy (%) and its confidence interval; Avg. is the arithmetic mean across datasets. Best point estimates are in **bold**.

Method	DeepSeek-R1-Distill-Qwen-7B					DeepSeek-R1-Distill-Llama-8B				
	AIME25	BRUMO25	MATH500	GPQA	Avg.	AIME25	BRUMO25	MATH500	GPQA	Avg.
Random	23.3 [10.0, 40.0]	23.3 [10.0, 40.0]	81.4 [78.0, 84.8]	49.5 [42.4, 56.6]	44.4	23.3 [10.0, 40.0]	16.7 [3.3, 30.0]	76.4 [72.6, 80.0]	47.5 [40.4, 54.5]	41.0
Self-Certainty	20.0 [6.7, 36.7]	10.0 [0.0, 20.0]	70.2 [66.2, 74.2]	52.0 [44.9, 58.6]	38.1	6.7 [0.0, 16.7]	10.0 [0.0, 23.3]	60.6 [56.4, 64.8]	46.5 [39.4, 53.5]	30.9
DeepConf	16.7 [3.3, 30.0]	13.3 [3.3, 26.7]	72.6 [68.6, 76.6]	54.0 [47.0, 60.6]	39.2	10.0 [0.0, 23.3]	16.7 [3.3, 30.0]	73.6 [69.6, 77.4]	53.0 [46.0, 60.1]	38.3
Self-Consistency	30.0 [13.3, 46.7]	46.7 [30.0, 63.3]	87.6 [84.6, 90.4]	52.5 [45.5, 59.6]	54.2	23.3 [10.0, 40.0]	26.7 [13.3, 43.3]	83.8 [80.6, 87.0]	54.0 [47.0, 61.1]	47.0
Low-Token-Entropy	26.7 [10.0, 43.3]	43.3 [26.7, 60.0]	86.4 [83.2, 89.2]	53.0 [46.5, 59.6]	52.4	16.7 [3.3, 30.0]	20.0 [6.7, 36.7]	83.0 [79.6, 86.2]	56.6 [49.5, 63.1]	44.1
UID-Global	30.0 [13.3, 46.7]	43.3 [26.7, 60.0]	90.6 [87.8, 93.0]	52.0 [44.9, 59.1]	54.0	30.0 [13.3, 46.7]	16.7 [3.3, 30.0]	83.6 [80.2, 86.8]	52.0 [44.9, 59.1]	45.6
UID-Local	30.0 [13.3, 46.7]	50.0 [33.3, 66.7]	90.0 [87.4, 92.6]	50.5 [43.4, 57.6]	55.1	33.3 [16.7, 50.0]	33.3 [16.7, 50.0]	84.0 [80.8, 87.2]	51.0 [43.9, 58.1]	50.4
LVE (Ours)	30.0 [13.3, 46.7]	50.0 [33.3, 66.7]	90.4 [87.8, 92.8]	55.6 [48.5, 62.6]	56.5	33.3 [16.7, 50.0]	36.7 [20.0, 53.3]	85.4 [82.2, 88.4]	54.5 [47.5, 61.1]	52.5
LogNorm-LVE (Ours)	30.0 [13.3, 46.7]	50.0 [33.3, 66.7]	92.2 [89.8, 94.4]	53.5 [46.5, 60.1]	56.4	33.3 [16.7, 50.0]	40.0 [23.3, 56.7]	85.0 [81.8, 88.0]	53.0 [46.0, 60.1]	52.8

and apply per-response min-max normalization $\widetilde{\text{ID}}(s_m)$. UID-Global is the variance of normalized step IDs,

$$\text{UID}_{\text{global}}(y) = \text{Var}\left(\widetilde{\text{ID}}(s_1), \dots, \widetilde{\text{ID}}(s_M)\right), \quad (19)$$

selected with $\arg \max$. UID-Local counts anomalous step-to-step changes with threshold $k = 2$:

$$\text{UID}_{\text{local}}(y) = \sum_{m=2}^M \mathbf{1}[\Delta_m > \mu_{\Delta} + k\sigma_{\Delta}] + \mathbf{1}[\Delta_m < \mu_{\Delta} - k\sigma_{\Delta}], \quad (20)$$

where $\Delta_m = \widetilde{\text{ID}}(s_m) - \widetilde{\text{ID}}(s_{m-1})$. Prior work notes that $k = 2$ or $k = 3$ are reasonable choices; in our experiments, we tested both $k = 2$ and $k = 3$ and found that $k = 2$ yields higher overall accuracy, so we use $k = 2$ for all comparisons. Selection uses $\arg \min$.

D. Additional Results and Analyses

This section complements the main experimental results with full quantitative tables and diagnostic analyses. We first report the complete $N = 10$ comparison with bootstrap confidence intervals, then provide per- N accuracy tables for all datasets. The section also includes candidate-level ROC curves beyond MATH500, model-diversity results on additional backbones, and supporting analyses for length control and velocity-mass concentration.

D.1. Main $N = 10$ Table with Bootstrap Confidence Intervals

Table 3 expands the main $N = 10$ comparison by reporting 95% bootstrap confidence intervals for every dataset and backbone. The Avg. column is the arithmetic mean of the four dataset-level point estimates. As in the main text, intervals on AIME25 and BRUMO25 are wider because these benchmarks contain 30 evaluation problems each, so several apparent differences fall within statistical uncertainty rather than reliable margins.

D.2. Accuracy Across Sampling Budgets

Tables 4 and 5 report the full Best-of- N selection accuracy across datasets and sample sizes. While the main text focuses on $N = 10$, these tables include $N \in \{3, 5, 10, 15\}$ to show how each selection rule behaves as the candidate pool grows. Overall, LogNorm-LVE remains competitive across N , with the clearest gains on MATH500; smaller datasets such as AIME25 and BRUMO25 exhibit more ties because their evaluation sets are limited in size.

Table 4. Full Best-of- N selection accuracy (%) on AIME25 and BRUMO25. Columns report $N \in \{3, 5, 10, 15\}$; each column highlights the best point estimate.

(a) AIME25								
Method	DeepSeek-R1-Qwen				DeepSeek-R1-Llama			
	$N = 3$	$N = 5$	$N = 10$	$N = 15$	$N = 3$	$N = 5$	$N = 10$	$N = 15$
Random	30.0	30.0	23.3	30.0	20.0	23.3	23.3	16.7
Self-Certainty	20.0	20.0	20.0	10.0	13.3	13.3	6.7	6.7
DeepConf	20.0	20.0	16.7	10.0	16.7	20.0	10.0	6.7
Self-Consistency	30.0	26.7	30.0	26.7	23.3	26.7	23.3	26.7
Low-Token-Entropy	30.0	30.0	26.7	23.3	16.7	23.3	16.7	20.0
UID-Global	30.0	30.0	30.0	33.3	23.3	23.3	30.0	26.7
UID-Local	30.0	30.0	30.0	30.0	23.3	26.7	33.3	26.7
LVE (Ours)	30.0	30.0	30.0	33.3	23.3	20.0	33.3	26.7
LogNorm-LVE (Ours)	30.0	30.0	30.0	33.3	23.3	23.3	33.3	26.7

(b) BRUMO25								
Method	DeepSeek-R1-Qwen				DeepSeek-R1-Llama			
	$N = 3$	$N = 5$	$N = 10$	$N = 15$	$N = 3$	$N = 5$	$N = 10$	$N = 15$
Random	33.3	26.7	23.3	23.3	20.0	20.0	16.7	13.3
Self-Certainty	16.7	13.3	10.0	10.0	16.7	10.0	10.0	3.3
DeepConf	20.0	16.7	13.3	10.0	16.7	16.7	16.7	10.0
Self-Consistency	33.3	33.3	46.7	36.7	23.3	23.3	26.7	33.3
Low-Token-Entropy	36.7	33.3	43.3	40.0	20.0	23.3	20.0	20.0
UID-Global	36.7	40.0	43.3	50.0	20.0	23.3	16.7	30.0
UID-Local	40.0	40.0	50.0	43.3	23.3	30.0	33.3	33.3
LVE (Ours)	43.3	46.7	50.0	43.3	23.3	30.0	36.7	33.3
LogNorm-LVE (Ours)	43.3	46.7	50.0	43.3	23.3	30.0	40.0	36.7

Table 5. Full Best-of- N selection accuracy (%) on GPQA and MATH500. Columns report $N \in \{3, 5, 10, 15\}$; each column highlights the best point estimate.

(a) GPQA								
Method	DeepSeek-R1-Qwen				DeepSeek-R1-Llama			
	$N = 3$	$N = 5$	$N = 10$	$N = 15$	$N = 3$	$N = 5$	$N = 10$	$N = 15$
Random	50.5	47.5	49.5	50.5	46.0	48.5	47.5	52.5
Self-Certainty	53.0	50.0	52.0	50.0	42.4	49.0	46.5	50.0
DeepConf	53.0	52.0	54.0	55.6	48.5	55.6	53.0	58.6
Self-Consistency	54.0	48.5	52.5	49.0	46.5	51.0	54.0	53.0
Low-Token-Entropy	54.5	49.0	53.0	53.5	50.0	54.0	56.6	60.1
UID-Global	51.5	50.5	52.0	52.0	47.5	54.0	52.0	52.0
UID-Local	54.0	46.5	50.5	52.0	51.0	50.5	51.0	51.5
LVE (Ours)	51.5	50.0	55.6	55.1	52.5	57.1	54.5	54.0
LogNorm-LVE (Ours)	53.0	50.0	53.5	55.6	51.5	56.6	53.0	55.6

(b) MATH500								
Method	DeepSeek-R1-Qwen				DeepSeek-R1-Llama			
	$N = 3$	$N = 5$	$N = 10$	$N = 15$	$N = 3$	$N = 5$	$N = 10$	$N = 15$
Random	82.6	82.8	81.4	81.2	76.0	77.2	76.4	75.2
Self-Certainty	77.0	75.4	70.2	70.4	70.0	66.4	60.6	59.8
DeepConf	77.6	77.0	72.6	72.0	74.2	72.2	73.6	73.8
Self-Consistency	84.2	87.0	87.6	88.0	78.0	83.8	83.8	83.4
Low-Token-Entropy	83.8	86.0	86.4	87.2	76.4	79.2	83.0	80.0
UID-Global	85.6	87.6	90.6	89.6	81.0	83.0	83.6	83.6
UID-Local	85.0	88.6	90.0	88.4	83.4	83.2	84.0	83.4
LVE (Ours)	87.4	88.8	90.4	90.4	83.6	84.0	85.4	83.8
LogNorm-LVE (Ours)	87.4	89.0	92.2	91.4	83.4	84.0	85.0	84.6

D.3. Candidate-Level ROC Across Datasets

We further report candidate-level ROC curves for AIME25, BRUMO25, and GPQA to assess score separability beyond MATH500. On AIME25 and BRUMO25, LVE/LogNorm-LVE typically yield curves closer to the top-left, indicating stronger ranking separability; on GPQA, curves are closer to the diagonal for most methods, suggesting weaker candidate-level separability on this benchmark.

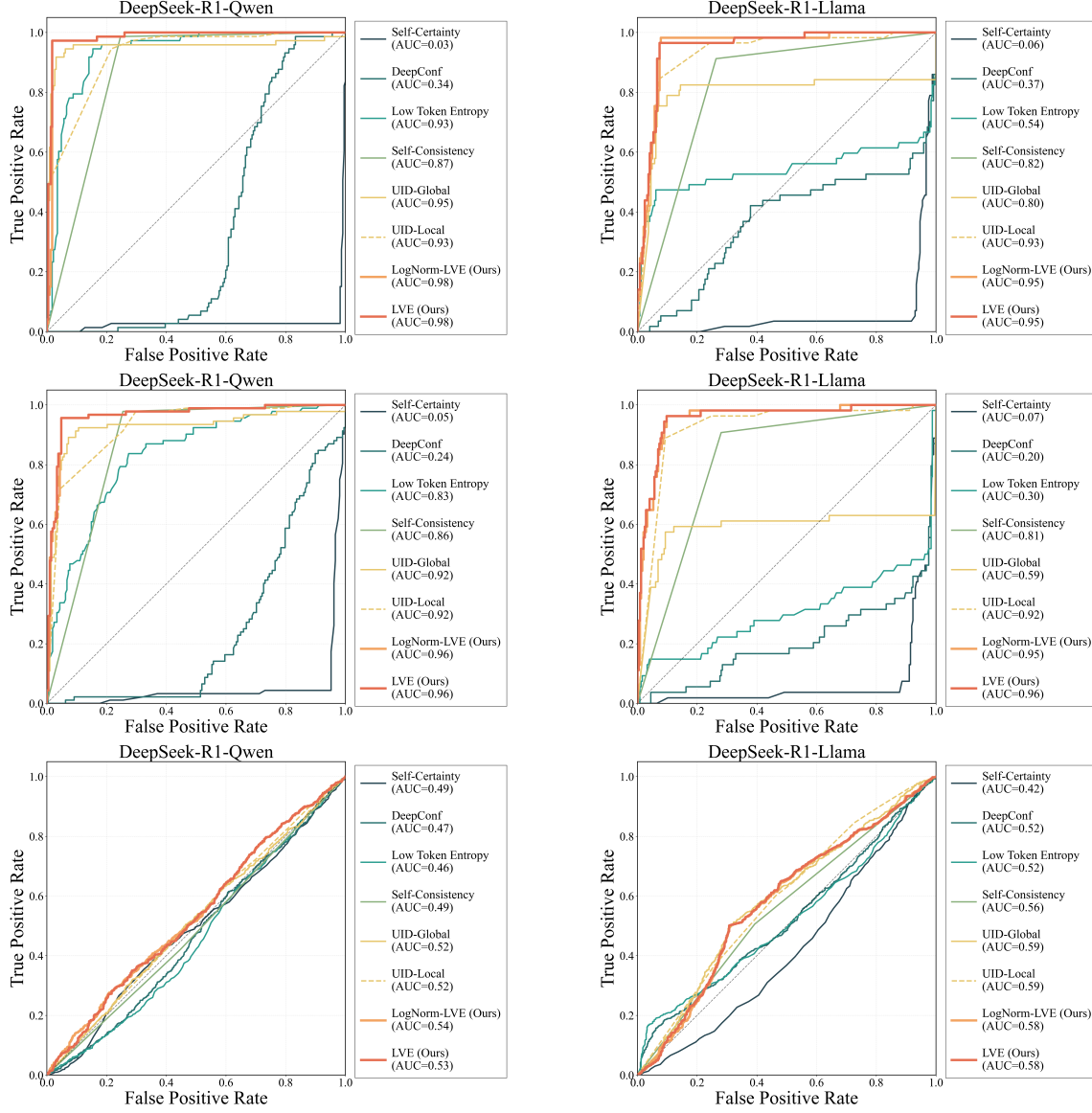


Figure 8. Candidate-level ROC curves across datasets (top: AIME25; middle: BRUMO25; bottom: GPQA). Curves closer to the top-left indicate better separability.

D.4. Model Diversity Analysis

To assess whether LVE is specific to the original DeepSeek-R1 distilled checkpoints, we evaluate two additional open-weight model families, Gemma-3-4B-It and Qwen3-8B, across the same four datasets and sample sizes. Across the resulting 32 settings, either LVE or LogNorm-LVE appears among the methods attaining one of the top three distinct point-estimate levels in 27 settings. Table 6 reports the full 32-setting model-diversity table with bootstrap confidence intervals. The pattern is mixed but informative: LVE variants remain competitive across architectures, with clearest advantages on MATH500 and weaker evidence on GPQA.

Table 6. Model diversity analysis across four datasets: Gemma-3-4B-It and Qwen3-8B. Each cell reports Pass@1 accuracy (%) with 95% bootstrap confidence intervals across sample sizes $N \in \{3, 5, 10, 15\}$. The highest point estimate in each row is shown in **bold**; ties are bolded jointly.

Model	N	Random	Self-Cert.	DeepConf	Self-Cons.	Low-Ent.	UID-G	UID-L	LVE	LogNorm
MATH500										
Gemma-3-4B-It	3	76.6 [72.8, 80.2]	78.4 [74.8, 82.0]	78.2 [74.6, 81.8]	78.2 [74.6, 81.8]	77.4 [73.8, 81.0]	77.0 [73.4, 80.6]	75.4 [71.6, 79.2]	78.8 [75.0, 82.4]	76.4 [72.6, 80.0]
	5	77.8 [74.2, 81.4]	76.8 [73.0, 80.4]	77.0 [73.4, 80.6]	79.2 [75.6, 82.6]	77.8 [74.2, 81.4]	77.2 [73.6, 80.8]	75.4 [71.6, 79.2]	77.8 [74.2, 81.4]	76.4 [72.8, 80.0]
	10	75.0 [71.2, 78.8]	77.4 [73.8, 81.0]	77.8 [74.0, 81.4]	80.4 [76.8, 83.8]	76.6 [73.0, 80.4]	77.6 [74.0, 81.2]	75.4 [71.6, 79.2]	79.4 [75.8, 83.0]	76.0 [72.2, 79.6]
	15	74.6 [70.8, 78.4]	77.0 [73.2, 80.6]	77.4 [73.8, 81.0]	81.0 [77.6, 84.4]	76.2 [72.4, 80.0]	78.0 [74.2, 81.6]	75.4 [71.6, 79.0]	78.4 [74.8, 82.0]	75.8 [72.0, 79.6]
Qwen3-8B	3	78.6 [75.0, 82.0]	82.2 [78.8, 85.4]	82.6 [79.2, 85.8]	81.8 [78.4, 85.2]	82.0 [78.6, 85.2]	75.8 [72.0, 79.6]	76.2 [72.4, 79.8]	82.4 [79.0, 85.6]	82.0 [78.6, 85.4]
	5	78.0 [74.4, 81.6]	83.8 [80.4, 87.0]	84.2 [81.0, 87.4]	83.8 [80.6, 87.0]	83.6 [80.4, 86.8]	75.2 [71.4, 79.0]	76.0 [72.2, 79.8]	84.2 [81.0, 87.2]	83.4 [80.0, 86.6]
	10	79.6 [76.0, 83.0]	86.4 [83.4, 89.4]	86.4 [83.2, 89.4]	86.0 [83.0, 89.0]	86.0 [83.0, 89.0]	75.2 [71.4, 79.0]	76.6 [72.8, 80.2]	86.8 [83.8, 89.8]	85.8 [82.6, 88.8]
	15	78.6 [75.0, 82.0]	87.2 [84.2, 90.0]	87.2 [84.2, 90.0]	86.6 [83.6, 89.6]	86.4 [83.2, 89.4]	75.2 [71.4, 79.0]	76.6 [72.8, 80.2]	86.8 [83.6, 89.6]	86.0 [82.8, 89.0]
GPQA										
Gemma-3-4B-It	3	28.3 [22.2, 34.8]	31.8 [25.3, 38.4]	32.8 [26.3, 39.4]	27.8 [21.7, 34.3]	33.3 [26.8, 39.9]	25.3 [19.7, 31.3]	26.8 [20.7, 32.8]	28.3 [22.2, 34.8]	29.8 [23.2, 36.4]
	5	24.7 [18.7, 30.8]	28.3 [22.2, 34.8]	27.3 [21.2, 33.3]	28.8 [22.7, 34.8]	26.3 [20.2, 32.3]	24.2 [18.7, 30.3]	26.8 [20.7, 32.8]	26.8 [20.7, 32.8]	27.8 [21.7, 34.3]
	10	25.3 [19.7, 31.3]	31.8 [25.3, 38.4]	30.8 [24.2, 37.4]	28.8 [22.7, 35.4]	29.8 [23.7, 36.4]	24.2 [18.7, 30.3]	26.8 [20.7, 32.8]	26.8 [20.7, 32.8]	29.3 [23.2, 35.9]
	15	26.3 [20.2, 32.3]	34.8 [28.3, 41.9]	32.3 [25.8, 38.9]	29.8 [23.7, 36.4]	33.3 [26.8, 39.9]	27.8 [21.7, 33.8]	26.8 [20.7, 32.8]	30.3 [23.7, 36.9]	32.3 [25.8, 38.9]
Qwen3-8B	3	63.1 [56.6, 69.7]	72.2 [65.7, 78.3]	71.7 [65.2, 77.8]	68.7 [62.1, 75.3]	70.2 [64.1, 76.3]	66.2 [59.6, 72.7]	68.7 [62.1, 75.3]	71.2 [64.6, 77.3]	71.2 [64.6, 77.3]
	5	65.2 [58.6, 71.7]	69.7 [63.1, 76.3]	69.2 [63.1, 75.3]	68.7 [62.1, 74.7]	69.7 [63.1, 75.8]	65.2 [58.6, 71.7]	69.2 [62.6, 75.8]	71.2 [64.6, 77.3]	70.7 [64.1, 76.8]
	10	65.2 [58.6, 71.7]	70.2 [63.6, 76.3]	71.2 [64.6, 77.3]	69.7 [63.1, 75.8]	70.7 [64.1, 76.8]	64.1 [57.6, 70.7]	68.7 [62.1, 75.3]	66.7 [60.1, 73.2]	69.2 [62.6, 75.8]
	15	66.7 [60.1, 73.2]	70.7 [64.1, 76.8]	72.2 [65.7, 78.3]	69.7 [63.1, 76.3]	71.2 [65.2, 77.3]	66.2 [59.6, 72.7]	67.7 [61.1, 74.2]	67.7 [61.1, 74.2]	69.2 [62.6, 75.3]
AIME25										
Gemma-3-4B-It	3	10.0 [0.0, 23.3]	16.7 [3.3, 30.0]	13.3 [3.3, 26.7]	10.0 [0.0, 20.0]	16.7 [3.3, 30.0]	13.3 [3.3, 26.7]	6.7 [0.0, 16.7]	13.3 [3.3, 26.7]	13.3 [3.3, 26.7]
	5	10.0 [0.0, 23.3]	23.3 [10.0, 40.0]	20.0 [6.7, 36.7]	13.3 [3.3, 26.7]	20.0 [6.7, 33.3]	13.3 [3.3, 26.7]	6.7 [0.0, 16.7]	16.7 [3.3, 30.0]	13.3 [3.3, 26.7]
	10	10.0 [0.0, 23.3]	23.3 [10.0, 40.0]	20.0 [6.7, 36.7]	16.7 [3.3, 30.0]	20.0 [6.7, 36.7]	13.3 [3.3, 26.7]	6.7 [0.0, 16.7]	6.7 [6.7, 33.4]	16.7 [3.3, 30.0]
	15	10.0 [0.0, 23.3]	20.0 [6.7, 33.3]	16.7 [3.3, 30.0]	13.3 [3.3, 26.7]	16.7 [3.3, 30.0]	13.3 [3.3, 26.7]	6.7 [0.0, 16.7]	20.0 [6.7, 36.7]	16.7 [3.3, 30.0]
Qwen3-8B	3	13.3 [3.3, 26.7]	20.0 [6.7, 36.7]	20.0 [6.7, 33.3]	13.3 [3.3, 26.7]	20.0 [6.7, 33.3]	10.0 [0.0, 23.3]	10.0 [0.0, 23.3]	20.0 [6.7, 33.3]	20.0 [6.7, 36.7]
	5	10.0 [0.0, 20.0]	20.0 [6.7, 33.3]	20.0 [6.7, 36.7]	13.3 [3.3, 26.7]	20.0 [6.7, 33.3]	10.0 [0.0, 23.3]	13.3 [3.3, 26.7]	20.0 [6.7, 33.3]	20.0 [6.7, 36.7]
	10	16.7 [3.3, 30.0]	23.3 [10.0, 40.0]	23.3 [10.0, 40.0]	23.3 [10.0, 40.0]	23.3 [10.0, 40.0]	16.7 [3.3, 30.0]	10.0 [0.0, 23.3]	23.3 [10.0, 40.0]	20.0 [6.7, 36.7]
	15	13.3 [3.3, 26.7]	26.7 [13.3, 43.3]	26.7 [13.3, 43.3]	23.3 [10.0, 40.0]	26.7 [13.3, 43.3]	16.7 [3.3, 30.0]	10.0 [0.0, 23.3]	26.7 [10.0, 43.3]	23.3 [10.0, 40.0]
BRUMO25										
Gemma-3-4B-It	3	20.0 [6.7, 33.3]	23.3 [10.0, 40.0]	20.0 [6.7, 36.7]	23.3 [10.0, 40.0]	20.0 [6.7, 36.7]	20.0 [6.7, 33.3]	20.0 [6.7, 33.3]	26.7 [13.3, 43.3]	26.7 [10.0, 43.3]
	5	23.3 [10.0, 40.0]	23.3 [10.0, 40.0]	23.3 [10.0, 40.0]	23.3 [10.0, 40.0]	20.0 [6.7, 33.3]	16.7 [3.3, 30.0]	20.0 [6.7, 33.3]	26.7 [13.3, 43.3]	23.3 [10.0, 40.0]
	10	20.0 [6.7, 33.3]	23.3 [10.0, 40.0]	23.3 [10.0, 40.0]	23.3 [10.0, 40.0]	26.7 [10.0, 43.3]	20.0 [6.7, 36.7]	20.0 [6.7, 33.3]	26.7 [13.3, 43.3]	23.3 [10.0, 40.0]
	15	16.7 [3.3, 30.0]	23.3 [10.0, 40.0]	23.3 [10.0, 40.0]	23.3 [10.0, 40.0]	23.3 [10.0, 40.0]	20.0 [6.7, 36.7]	20.0 [6.7, 36.7]	26.7 [13.3, 43.3]	26.7 [10.0, 43.3]
Qwen3-8B	3	23.3 [10.0, 40.0]	23.3 [10.0, 40.0]	20.0 [6.7, 36.7]	20.0 [6.7, 33.3]	23.3 [10.0, 40.0]	16.7 [3.3, 30.0]	16.7 [3.3, 30.0]	23.3 [10.0, 40.0]	20.0 [6.7, 36.7]
	5	20.0 [6.7, 33.3]	23.3 [10.0, 40.0]	23.3 [10.0, 40.0]	20.0 [6.7, 36.7]	23.3 [10.0, 40.0]	13.3 [3.3, 26.7]	13.3 [3.3, 26.7]	23.3 [10.0, 40.0]	26.7 [13.3, 43.3]
	10	16.7 [3.3, 30.0]	26.7 [13.3, 43.3]	23.3 [10.0, 40.0]	23.3 [10.0, 40.0]	20.0 [6.7, 33.3]	13.3 [3.3, 26.7]	13.3 [3.3, 26.7]	20.0 [6.7, 33.4]	20.0 [6.7, 36.7]
	15	16.7 [3.3, 30.0]	33.3 [16.7, 50.0]	30.0 [13.3, 46.7]	26.7 [13.3, 43.3]	26.7 [13.3, 43.3]	13.3 [3.3, 26.7]	13.3 [3.3, 26.7]	23.3 [10.0, 40.0]	23.3 [10.0, 40.0]

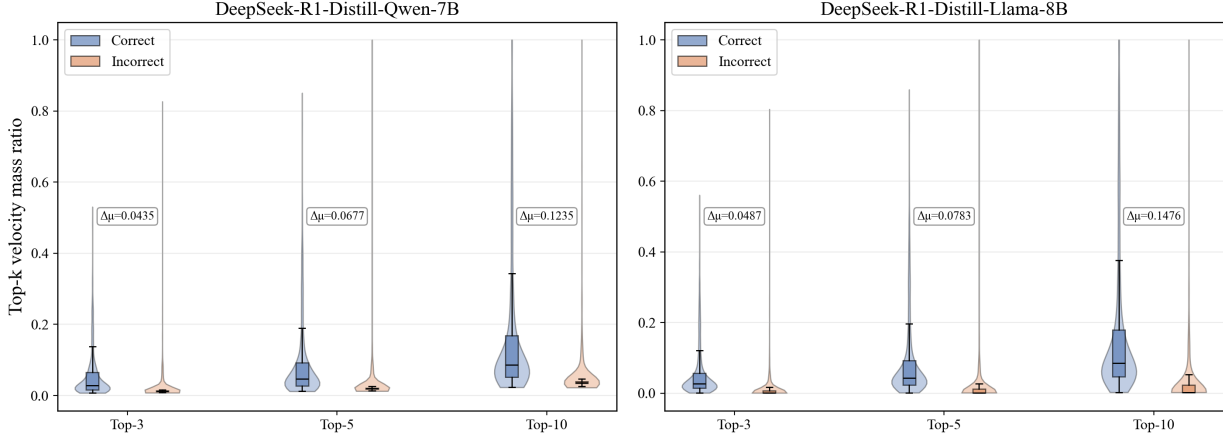


Figure 9. Top- k velocity mass ratios on MATH500. Correct candidates exhibit higher temporal concentration of latent update mass than incorrect candidates for $k \in \{3, 5, 10\}$, with the separation increasing for larger k in both evaluated DeepSeek-R1 distilled backbones.

D.5. Length-Controlled Analysis

Raw LVE is partially correlated with response length, motivating an explicit length-controlled analysis. On MATH500 with DeepSeek-R1-Distill-Qwen-7B, raw LVE has Pearson correlation 0.384 with token length and 0.389 with log-token-length. Accordingly, LogNorm-LVE serves as the preferred length-normalized variant when candidate lengths differ.

At the same time, length does not fully explain LVE’s discriminative behavior. We restrict attention to same-problem correct/incorrect candidate pairs from the same sampling setting whose lengths differ by at most $\pm 10\%$. Under this explicit control, raw LVE remains informative: the mean problem-level pairwise AUC is 0.668, 63.0% of problems have AUC above 0.5, and the effect is significant under a Wilcoxon test ($p = 1.99 \times 10^{-7}$). After residualizing LVE by removing the component linearly explained by length and sampling scale, the result is essentially unchanged (mean AUC 0.667, 63.6% of problems above 0.5, $p = 3.58 \times 10^{-7}$).

We also evaluate whether LVE adds information beyond length in a predictive setting. Using only length to predict candidate correctness yields 5-fold GroupKFold AUROC 0.517, with folds grouped by problem so that candidates from the same problem never appear in both train and test. Adding raw LVE increases AUROC to 0.626, a gain of +0.109. Finally, direct selection by minimum LVE and shortest length chooses different trajectories in 70.1% of problem groups. These controls support the interpretation used in the main text: raw LVE is length-related but not reducible to length alone.

D.6. Top- k Velocity Mass Concentration

We further test the temporal-concentration hypothesis with a Top- k velocity mass ratio: for each candidate, we sort per-step latent velocities and measure the fraction of total update mass contained in the largest k steps. Figure 9 shows that correct candidates tend to allocate more velocity mass to their largest updates than incorrect candidates, across both DeepSeek-R1-Distill-Qwen-7B and DeepSeek-R1-Distill-Llama-8B. This supports the empirical “dwell-and-jump” interpretation used by LVE while keeping the biological analogy as a motivating prior.

D.7. Layer Position Robustness

We evaluate the robustness of LogNorm-LVE with respect to the choice of transformer layer by applying the selection criterion to hidden states extracted from different layers of the backbone models. Specifically, for each layer, we compute the LogNorm-LVE score based on the corresponding hidden representations and report the resulting Pass@1 selection accuracy.

Figure 10 shows the layer-wise performance on BRUMO25 ($N = 10$) and AIME25 ($N = 15$) for both DeepSeek-R1-Qwen and DeepSeek-R1-Llama. Across all settings, LogNorm-LVE exhibits stable performance over a wide range of layers, with only minor fluctuations in early or middle layers for certain models. In these settings, the method remains above self-consistency across many layer positions.

These results suggest that LogNorm-LVE does not rely on a single specific layer and is relatively stable across representation

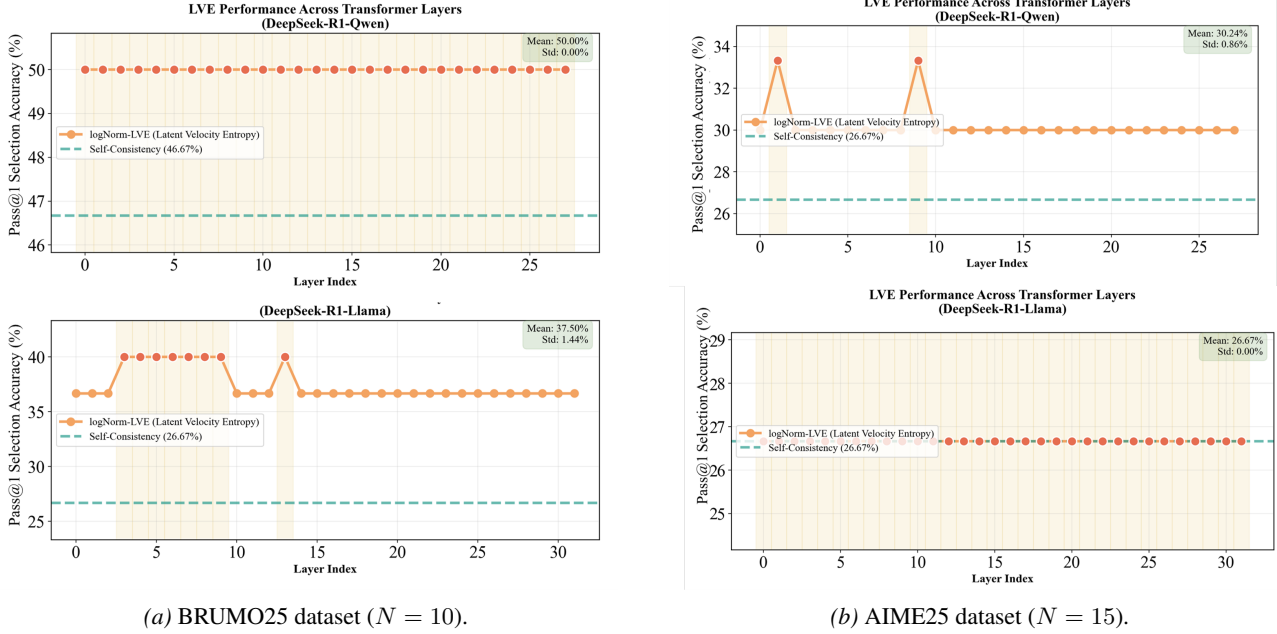


Figure 10. Pass@1 selection accuracy across transformer layers for different datasets and models.

depths, reducing the need for extensive layer-specific tuning.

D.8. Error Dynamics in Reasoning Trajectories

Our analysis indicates that erroneous reasoning trajectories exhibit distinct patterns. We identify two representative regimes.

Single-point decision (Figure 11) errors are characterized by a hard, discrete choice made at a critical token segment, where the model incorrectly selects a directional label (“west” instead of “east”: “Starting from north, rotating 90 degrees to the right would face the skater towards the west.”). Such errors can be hard to recover from within the sampled trajectory: once the choice is made, subsequent reasoning proceeds in a self-consistent but incorrect manner. Dynamically, this regime manifests as a sharp spike in the non-cumulative probability mass p_t and a corresponding kink in the CDF, indicating a sudden branch selection in the hidden-state trajectory. This behavior differs from legitimate logical step transitions in correct reasoning, which typically span multiple tokens and induce smoother, wider probability redistribution rather than narrow peaks.

In contrast, distributed or progressive errors (Figure 12) arise from sustained misalignment throughout the reasoning process, such as algebraic mis-expansion, diagram misinterpretation, or incomplete enumeration. These errors do not correspond to a single decisive token; instead, small errors accumulate across steps. As a result, probability mass remains diffusely distributed over tokens, yielding flat p_t profiles and near-linear CDFs. While both correct and incorrect trajectories in this regime may appear smooth, erroneous trajectories keep their update mass broadly distributed rather than concentrating around identifiable transition points.

Overall, these results suggest that correct and incorrect reasoning trajectories are not linearly separable at the token level, but are distinguishable by their temporal diffusion geometry.

D.9. Distribution of Internal Dynamics Metrics

To complement the main analysis, we report the empirical distributions of three velocity-based internal dynamics metrics computed over all generated samples.

Specifically, for each response, we extract hidden-state trajectories at a fixed intermediate layer (layer 15, DeepSeek-R1-Distill-Qwen-7B on MATH500) and compute: (i) the cumulative latent velocity (VSum), defined as the sum of ℓ_2 norms of consecutive hidden-state differences; (ii) the length-normalized average velocity (VAvg), obtained by normalizing VSum

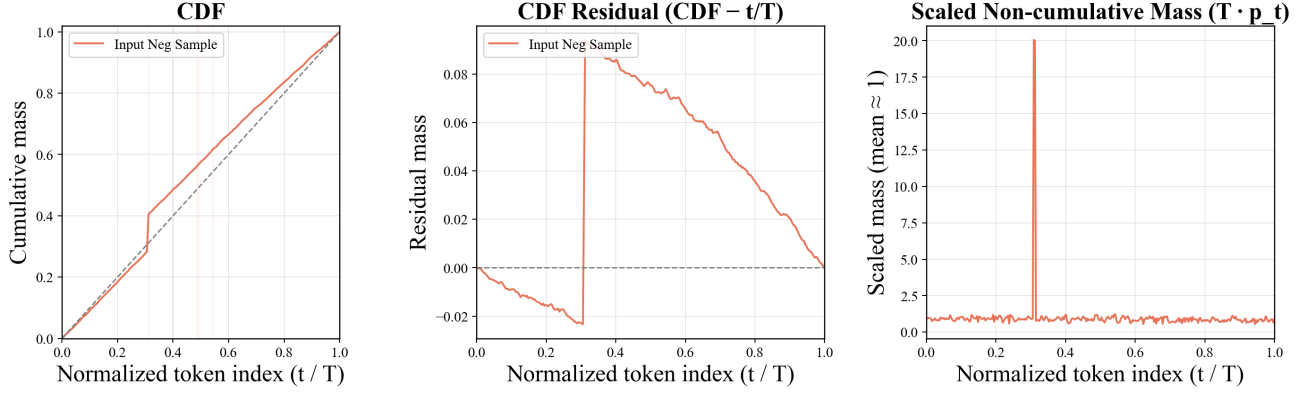


Figure 11. Velocity-based internal dynamics of a representative *single-point decision error* trajectory (DeepSeek-R1-Distill-Qwen-7B on MATH500). Left: the CDF exhibits a clear kink around a specific normalized token index, deviating sharply from the linear baseline. Middle: the CDF residual shows an abrupt jump followed by a monotonic relaxation, indicating a discrete branch selection in the hidden-state trajectory. Right: the scaled non-cumulative mass $T \cdot p_t$ displays a pronounced spike, revealing a strong concentration of probability mass at a narrow token segment.

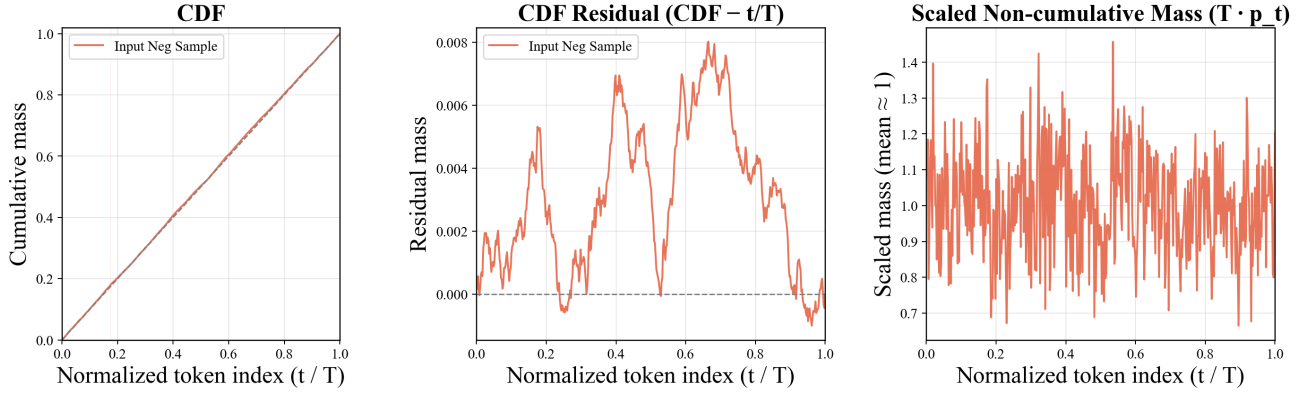


Figure 12. Velocity-based internal dynamics of a representative *distributed error* trajectory (DeepSeek-R1-Distill-Qwen-7B on MATH500). Left: the cumulative distribution function (CDF) of normalized velocity mass closely follows the linear baseline t/T , indicating approximately uniform diffusion over the reasoning process. Middle: the length-free CDF residual ($\text{CDF} - t/T$) exhibits low-amplitude, slowly varying deviations without sharp transitions. Right: the scaled non-cumulative mass $T \cdot p_t$ fluctuates around its mean across tokens, with no dominant peaks.

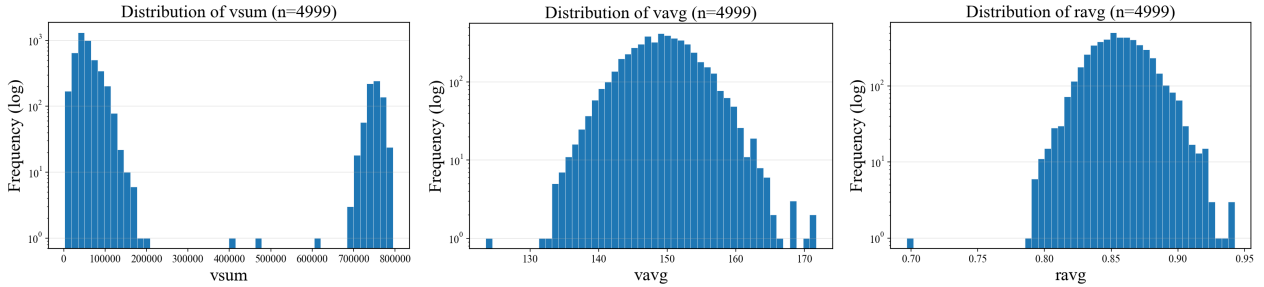


Figure 13. Distributions of velocity-based internal dynamics metrics over all generated samples. From left to right: cumulative latent velocity (VSum), length-normalized average velocity (VAvg), and relative average velocity (RAvg). All histograms are plotted with logarithmic frequency scaling.

by the response length; and (iii) the relative average velocity (RAvg), defined as the mean ratio between latent velocity and the corresponding hidden-state norm. For each metric, we visualize its marginal distribution using a log-scaled histogram.